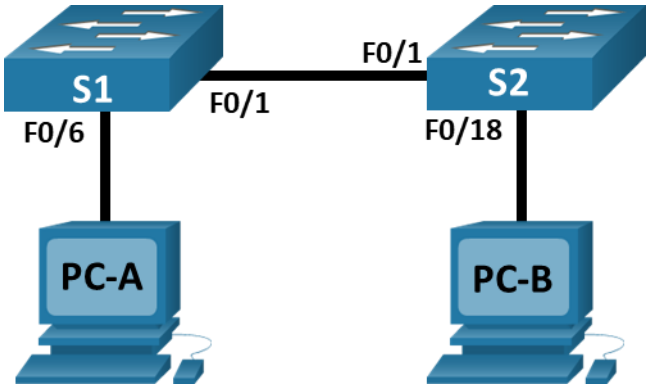
Lab - View the Switch MAC Address Table

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
S1	VLAN 1	192.168.1.11	255.255.255.0
S2	VLAN 1	192.168.1.12	255.255.255.0
PC-A	NIC	192.168.1.1	255.255.255.0
PC-B	NIC	192.168.1.2	255.255.255.0

Objectives

Part 1: Build and Configure the Network

Part 2: Examine the Switch MAC Address Table

Background / Scenario

The purpose of a Layer 2 LAN switch is to deliver Ethernet frames to host devices on the local network. The switch records host MAC addresses that are visible on the network, and maps those MAC addresses to its own Ethernet switch ports. This process is called building the MAC address table. When a switch receives a frame from a PC, it examines the frame's source and destination MAC addresses. The source MAC address is recorded and mapped to the switch port from which it arrived. Then the destination MAC address is looked up in the MAC address table. If the destination MAC address is a known address, then the frame is forwarded out of the corresponding switch port associated with that MAC address. If the MAC address is unknown, then the frame is broadcasted out of all switch ports, except the one from which it came. It is important to observe and understand the function of a switch and how it delivers data on the network. The way a switch operates has implications for network administrators whose job it is to ensure secure and consistent network communication.

Switches are used to interconnect and deliver information to computers on local area networks. Switches deliver Ethernet frames to host devices identified by network interface card MAC addresses.

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In Part 1, you will build a multi-switch topology with a trunk linking the two switches. In Part 2, you will ping various devices and observe how the two switches build their MAC address tables.

Note: The switches used are Cisco Catalyst 2960s with Cisco IOS Release 15.2(2) (lanbasek9 image). Other switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and output produced might vary from what is shown in the labs.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 2 Switches (Cisco 2960 with Cisco IOS Release 15.2(2) lanbasek9 image or comparable)
- 2 PCs (Windows with terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Note: The Fast Ethernet interfaces on Cisco 2960 switches are autosensing and an Ethernet straight-through cable may be used between switches S1 and S2. If using another model Cisco switch, it may be necessary to use an Ethernet crossover cable.

Instructions

Part 1: Build and Configure the Network

Step 1: Cable the network according to the topology.

Step 2: Configure PC hosts.

Step 3: Initialize and reload switches as necessary.

Step 4: Configure basic settings for each switch.

a. Configure device name as shown in the topology.

b. Configure IP address as listed in Addressing Table.

c. Assign **cisco** as the console and vty passwords.

d. Assign **class** as the privileged EXEC password.

Part 2: Examine the Switch MAC Address Table

A switch learns MAC addresses and builds the MAC address table, as network devices initiate communication on the network.

Step 1: Record network device MAC addresses.

a. Open a command prompt on PC-A and PC-B and type **ipconfig /all**.

What are the Ethernet adapter physical addresses?

PC-A MAC Address:

0002.1735.0A2E

PC-B MAC Address:

0001.965C.724B

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b. Console into switch S1 and S2 and type the **show interface F0/1** command on each switch.

On the second line of command output, what is the hardware addresses (or burned-in address [bia])?

S1 Fast Ethernet 0/1 MAC Address:

0090.0c2d.4b01

S2 Fast Ethernet 0/1 MAC Address:

0006.2a75.3301

Step 2: Display the switch MAC address table.

Console into switch S2 and view the MAC address table, both before and after running network communication tests with ping.

a. Establish a console connection to S2 and enter privileged EXEC mode.

b. In privileged EXEC mode, type the **show mac address-table** command and press Enter.

```
S2# show mac address-table
```

Even though there has been no network communication initiated across the network (i.e., no use of ping), it is possible that the switch has learned MAC addresses from its connection to the PC and the other switch.

Are there any MAC addresses recorded in the MAC address table?

Yes, Even before I ran any pings, S2 already had a couple of dynamic entries, because the switches exchange STP BPDUs and CDP messages as soon as the trunk comes up.

What MAC addresses are recorded in the table? To which switch ports are they mapped and to which devices do they belong? Ignore MAC addresses that are mapped to the CPU.

Vlan	Mac Address	Type	Ports
1	0001.965c.724b	DYNAMIC	Fa0/1
1	0002.1735.0a2e	DYNAMIC	Fa0/2
1	0090.0c2d.4b02	DYNAMIC	Fa0/2

If you had not previously recorded MAC addresses of network devices in Step 1, how could you tell which devices the MAC addresses belong to, using only the output from the **show mac address-table** command? Does it work in all scenarios?

You can partly work it out from the port each MAC is mapped to. If you know which cable/port connects to which device (F0/18 goes to PC-B, F0/1 goes to S1), you can infer the device. But it does not work in all scenarios: a trunk port will show many MACs from many devices, a port connected to a hub or another switch can show multiple MACs, and you can't identify devices behind a trunk or wireless AP from the table alone. You'd also need to correlate with the topology or the OUI.

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Step 3: Clear the S2 MAC address table and display the MAC address table again.

a. In privileged EXEC mode, type the **clear mac address-table dynamic** command and press **Enter**.

```
S2# clear mac address-table dynamic
```

b. Quickly type the **show mac address-table** command again.

Does the MAC address table have any addresses in it for VLAN 1? Are there other MAC addresses listed?

Right after clear mac address-table dynamic, there are no dynamic VLAN 1 entries. The only addresses left are the static entries mapped to the CPU (e.g. 0100.0ccc.cccc, 0180.c200.0001–0010, and ffff.ffff.ffff), which are control/reserved addresses and are not cleared.

Wait 10 seconds, type the **show mac address-table** command, and press Enter. Are there new addresses in the MAC address table?

Yes. The dynamic entries reappear — S1's MAC (0090.0c2d.4b01) shows up again on F0/1 because the switches keep sending STP BPDUs every 2 seconds. PC-B's MAC may also reappear if it sends any traffic. The table rebuilds itself automatically

Step 4: From PC-B, ping the devices on the network and observe the switch MAC address table.

a. From PC-B, open a command prompt and type **arp -a**.

Not including multicast or broadcast addresses, how many device IP-to-MAC address pairs have been learned by ARP?

None at first — 0 entries. PC-B had not yet talked to any other device, so its ARP cache was empty.

b. From the PC-B command prompt, ping PC-A, S1, and S2.

Did all devices have successful replies? If not, check your cabling and IP configurations.

Yes. Pings to PC-A (192.168.1.1), S1 (192.168.1.11), and S2 (192.168.1.12) all returned replies.

c. From a console connection to S2, enter the **show mac address-table** command.

Has the switch added additional MAC addresses to the MAC address table? If so, which addresses and devices?

0001.965c.724b (PC-B), 0002.1735.0a2e (PC-A), 0090.0c2d.4b02 (PC-A)

From PC-B, open a command prompt and retype **arp -a**.

Does the PC-B ARP cache have additional entries for all network devices that were sent pings?

Yes

Internet Address	Physical Address	Type
192.168.1.1	0002.1735.0a2e	dynamic
192.168.1.11	000d.bdca.2cad	dynamic
192.168.1.12	0090.2105.2a24	dynamic

Reflection Question

On Ethernet networks, data is delivered to devices by their MAC addresses. For this to happen, switches and PCs dynamically build ARP caches and MAC address tables. With only a few computers on the network this process seems fairly easy. What might be some of the challenges on larger networks?

On a large network the MAC address table can grow very large, and switches have a limited CAM table size — if it overflows, the switch starts flooding unknown unicast frames, which hurts performance. Broadcast and multicast traffic (ARP, DHCP, STP) also scales badly, which is why large networks are segmented into VLANs and subnets. Aging timers mean entries constantly change, making troubleshooting harder. There are also security concerns: MAC spoofing, CAM table overflow attacks, and rogue devices. And from an admin point of view, correlating thousands of MAC entries to physical devices and ports becomes impractical without additional tools like NMS, LLDP/CDP neighbor data, and port-security features.

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